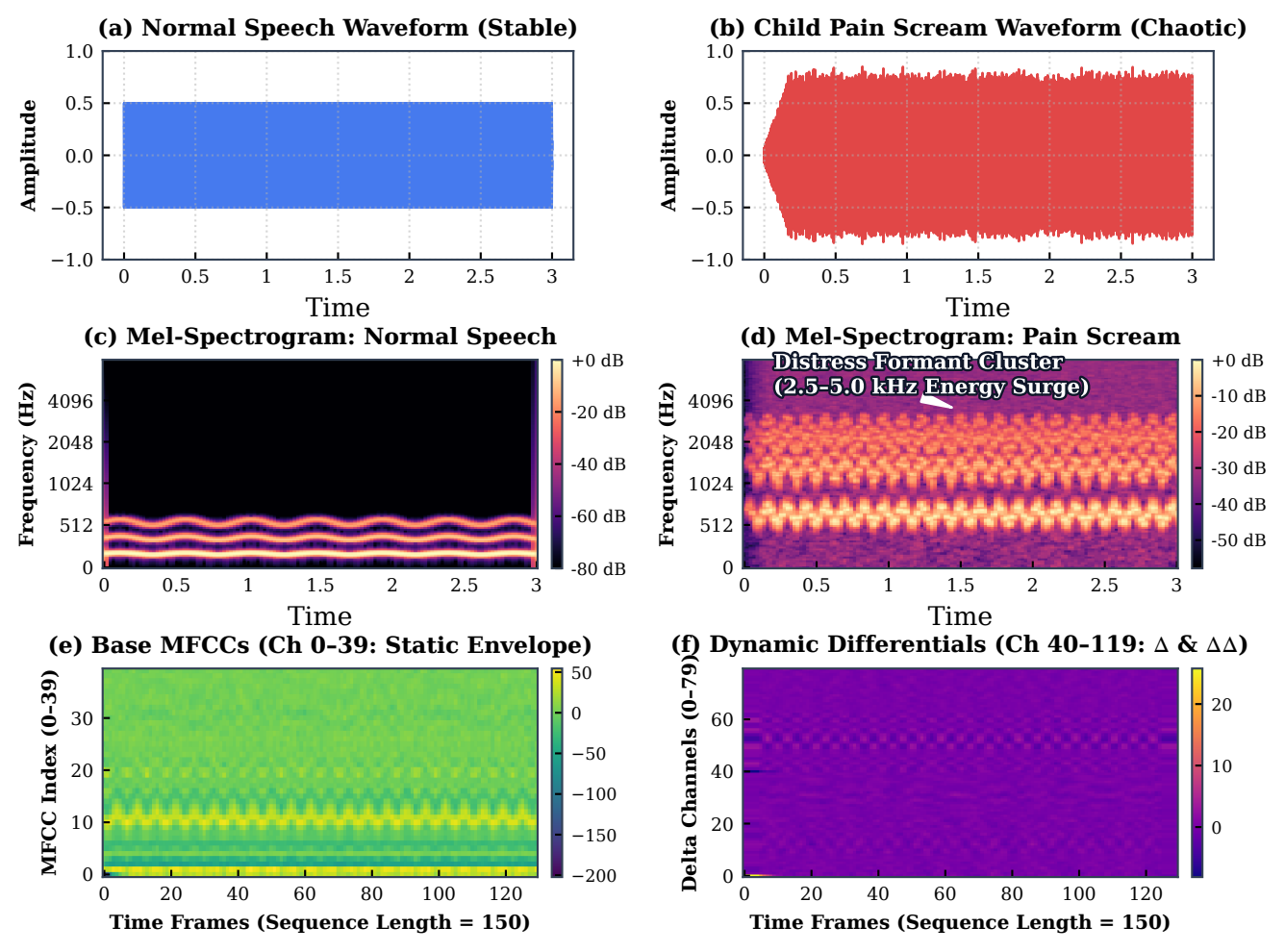

(a) Normal Speech Waveform (Stable)
Amplitude
Time
(b) Child Pain Scream Waveform (Chaotic)
Amplitude
Time
(c) Mel-Spectrogram: Normal Speech
Frequency (Hz)
Time
+0 dB
-20 dB
-40 dB
-60 dB
-80 dB
(d) Mel-Spectrogram: Pain Scream
Distress Formant Cluster
(2.5–5.0 kHz Energy Surge)
Frequency (Hz)
Time
+0 dB
-10 dB
-20 dB
-30 dB
-40 dB
-50 dB
(e) Base MFCCs (Ch 0–39: Static Envelope)
MFCC Index (0–39)
Time Frames (Sequence Length = 150)
(f) Dynamic Differentials (Ch 40–119: Δ & ΔΔ)
Delta Channels (0–79)
Time Frames (Sequence Length = 150)